



Detect Suspicious Network Behavior

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AI for Cybersecurity

https://colab.research.google.com/github/omarmkhallati/AI_For_Cybersecurity-Mkhallati/blob/main/Mohammed_Omar_Mkhallati-Suspicious_Network_Behavior.ipynb

Submitted Monday, September 14, 2026

Part A – Clustering

Since the data we are using is not labelled, i.e., we do not know if it is malicious or not, this is an unsupervised learning problem. We only have raw traffic attributes (timestamps, source/destination IPs, protocol, packet length), and the goal is to discover patterns and group similar traffic together rather than predict a known outcome. DBSCAN is the most suitable algorithm for this dataset. It does not require the number of clusters to be chosen in advance, which matters here since we do not know how many types of traffic behavior exist, and it groups sources based on density instead of forcing every point into a cluster, so sources that do not fit any dense group are labelled as noise instead of being pulled into a cluster they do not belong to. The records DBSCAN marks as noise (cluster label -1) are the abnormal sources, since they do not fit into any dense group of similar traffic behavior, and running this on the dataset flagged 23 out of 337 sources, about 7 percent, as noise, falling into a few different types of abnormal behavior rather than one single pattern: some sources send an unusually high volume of traffic or sustain a very high packet rate compared to the rest of the network, others use an unusually high number of different protocols while most sources only ever use one, and a few sources are also very low activity or come from addresses that are not typical host addresses, which is worth flagging for review as well.

Part B – Cybersecurity Analysis

Zero-day attacks exploit vulnerabilities that are not yet known, so there is no existing signature or label to detect them with, and signature-based or supervised methods can only catch attacks that have already been seen before and used to build a model, whereas anomaly detection instead looks for traffic that deviates from what is normal for the network, allowing it to flag a

zero-day attack the first time it happens purely because it behaves differently, without needing to have seen that specific attack before; this is where the comparison between K-Means and DBSCAN becomes relevant, since K-Means forces every source into one of a fixed number of clusters even when it does not really belong anywhere, which in this dataset showed up as the most extreme source being pulled into its own one-member cluster rather than being explicitly flagged as an anomaly, while DBSCAN instead groups sources by density and explicitly labels anything that does not belong to a dense group as noise, giving a direct anomaly flag rather than something that has to be inferred afterwards, and also not requiring the number of behaviour types to be guessed in advance, which matters for cybersecurity since new traffic patterns and new attacks are exactly what the analysis is trying to catch and not something that can be known ahead of time, making DBSCAN the more suitable algorithm for cybersecurity anomaly detection while K-Means is better suited to segmenting traffic into broad, already-expected categories.